

# VENKATA SAI AKHIL ANGA

716-650-8315 | [venkatasaiakhilanga@gmail.com](mailto:venkatasaiakhilanga@gmail.com) | [LinkedIn](#) | [GitHub](#) | Buffalo, NY

## PROFESSIONAL SUMMARY

Full Stack Software Engineer with over 4 years of experience designing scalable backend systems, RESTful APIs, microservices, and cloud-native applications using Node.js, Java, Spring Boot, Angular, and AWS. Experienced in distributed systems, performance optimization, CI/CD, and database engineering. Holds an MS in Data Science from the University at Buffalo.

## TECHNICAL SKILLS

**Languages:** Java, Python, JavaScript, TypeScript, SQL

**Backend:** Node.js, Spring Boot, Express.js

**Frontend:** Angular, TypeScript, JavaScript, HTML, CSS, Bootstrap

**Cloud & DevOps:** AWS (Lambda, API Gateway, S3, EC2, DynamoDB), Docker, Jenkins, CI/CD Pipelines

**Databases:** MySQL, PostgreSQL, Redis, DynamoDB, Cassandra

**Distributed Systems:** Apache Kafka, RabbitMQ, Microservices

**AI & Data:** Machine Learning, GenAI, Predictive Modeling, dbt, Streamlit, Tableau

**Core Concepts:** Microservices, RESTful APIs, System Design, OOP, Agile, Test Automation

## PROFESSIONAL EXPERIENCE

### Machinery Monitoring Systems LLC

Buffalo, NY

Software & Machine Learning Engineer (Internship)

Jan 2026 – May 2026

- Engineered a fault-severity detection prototype that cut manual inspection time by **30%** by processing real-time vibration data through **Python and Jenkins CI/CD pipelines** to automatically classify machinery faults before they caused downtime.
- Identified the optimal fault classification model by training and benchmarking 5 algorithms (Random Forest, XGBoost, LightGBM, MiniRocket, and CNN) using **FFT-based feature extraction**; selected **MiniRocket**, which achieved **99% classification accuracy** and was integrated into the final system.
- Accelerated model development by creating a labeled dataset of 10,000+ triaxial vibration samples in **MySQL**, enabling reliable training across multiple fault-severity classes.
- Reduced dashboard data retrieval latency by **30%** by developing real-time monitoring dashboards in Python backed by **Redis caching**, surfacing vibration trends, fault confidence scores, and severity levels for operators.
- Secured full stakeholder sign-off by leading technical demos and model walkthroughs with clients and faculty advisors, presenting results drawn from **MySQL and Cassandra** and clearly mapping model performance to business outcomes.

### Trivedi Advanced Technologies LLC

India

Full Stack Web Developer

Sep 2023 – Nov 2024

- Drove a **15% reduction in support tickets**, **30% improvement in API response times**, and a **33% growth in active users** by delivering full-stack product features using **Angular, TypeScript, Node.js, and CSS**.
- Improved production reliability and reduced API-related defects by **20%** by building and maintaining **RESTful APIs with Node.js and Express**, enforcing unit test coverage throughout the codebase.
- Reduced query execution times by 25% and eliminated data retrieval bottlenecks by integrating frontend services through **AWS API Gateway** and optimizing **SQL** and **DynamoDB** access patterns.
- Cut production issue investigation time by **25%** by implementing **Splunk** log monitoring dashboards across services, giving the team centralized visibility into system health.
- Delivered all sprint commitments on schedule while prototyping a **Spring Boot** service that validated a path to scaling the backend, which was adopted as the foundation for the next architecture phase.

Senior Software Engineer

Jan 2022 – Sep 2023

- Reduced database load by **35%** and improved response times for high-traffic services by architecting a microservices system using **Docker, Kafka, Redis, and RabbitMQ**, replacing a monolithic backend that was struggling under peak load.
- Improved deployment efficiency and cut infrastructure overhead by migrating backend services to **AWS Lambda, API Gateway, and S3**, while managing data storage across DynamoDB, Cassandra, and SQL based on each service's access patterns.
- Achieved consistent cross-browser and cross-device behavior by building the UI layer in **Angular, TypeScript, HTML, and Bootstrap**, which reduced front-end code duplication and kept page load times within acceptable limits.
- Reduced post-release defects by 20% through structured code reviews, cross-team collaboration, and early QA involvement throughout sprint execution.

Software Engineer

Jun 2020 – Jan 2022

- Improved backend service reliability by developing **RESTful APIs** in **Node.js** and **TypeScript**, supported by Splunk monitoring and Git-based release management.
- Reduced manual regression testing effort by **~40%** by building a **Selenium test automation suite in Java** with test cases tracked in TestRail, enabling faster and more consistent release validation.
- Shortened production incident resolution time by owning end-to-end debugging using **Splunk logs and Git history**, progressing from flagging issues to independently resolving them.

PROJECTS

Microservices Backend System Node.js · Docker · Kafka · Redis · AWS Lambda

May 2025

- Built a distributed backend with separate services communicating over **Kafka**, caching hot data in **Redis**, and running serverless jobs on **AWS Lambda**; the system handled traffic spikes without any single point of failure.

Healthcare Admissions Analytics System PostgreSQL · dbt · Python · Docker · Streamlit · SQL

Nov 2025 – Dec 2025

- Designed a healthcare data platform from the ground up, building a **PostgreSQL OLTP layer (3NF)** for raw data and a **dbt OLAP star schema** on top of it, using window functions and aggregations to answer questions like hospital revenue rankings and patient readmission patterns.
- Used EXPLAIN ANALYZE and indexing to cut query times by **~23%**; packaged the whole thing with Docker and built a **Streamlit** dashboard so non-technical users could explore admissions and billing data themselves.

Obesity Level Prediction & Lifestyle Pattern Analysis R · Random Forest · K-Means · Machine Learning

Oct 2025 – Dec 2025

- Used R to classify obesity levels from a real-world health dataset, training **Logistic Regression and Random Forest** models and evaluating them with balanced accuracy and macro F1-score to handle the class imbalance in the data.
- Ran **K-Means clustering** on the same dataset to find natural lifestyle groupings, using silhouette scores and Adjusted Rand Index (ARI) to validate the clusters, finding patterns that mapped well to real-world obesity categories.

EDUCATION

University at Buffalo, The State University of New York Jan 2025 – May 2026

Master of Science, Engineering Science – Data Science Buffalo, NY

Coursework: Distributed Systems, Cloud Computing, Machine Learning, Data Science

Sagi Ramakrishnam Raju Engineering College, Andhra University Aug 2017 – May 2020

Bachelor of Technology, Computer Science Engineering Visakhapatnam, India